Hammad Ahmad

AI/ML Researcher | Trustworthy NLP · Retrieval-Augmented Generation · Knowledge Graphs

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LinkedIn | GitHub | Portfolio | ORCID

RESEARCH PROFILE

AI/ML researcher with an MSc (Merit) in Applied AI and a first-author peer-reviewed publication (Springer). My work centres on reliable, transparent language systems: retrieval-augmented generation grounded in knowledge graphs to reduce hallucination, with rigorous benchmarking, alongside predictive modelling on high-dimensional real-world health data. I am seeking a PhD to develop this into rigorous research on trustworthy, knowledge-grounded NLP.

RESEARCH INTERESTS

Knowledge-grounded and retrieval-augmented NLP; hallucination reduction through structured grounding; evaluation frameworks for RAG systems; interpretable machine learning on high-dimensional health data.

EDUCATION

University of Bradford — Sep 2024 - Sep 2025

MSc, Applied Artificial Intelligence and Data Analytics (Merit) — Bradford, UK

- **Dissertation**: FinLaw-UK: A Graph-Augmented Retrieval Chatbot for Reliable and Transparent UK Financial Regulation. Supervisors: Dr Tillal Eldabi and Dr Irfan Mehmood. Detailed under Research Experience below.
- Selected modules: Artificial Intelligence and Data Science (79), Business Data Analytics (79), Responsible AI: Ethics, Law and Governance (75).

COMSATS University Islamabad — Sep 2020 - Jul 2024

BS, Bioinformatics — Islamabad, Pakistan

- **Thesis**: AI-Assisted Analysis and Prediction of At-Risk Diabetic Individuals (final year research project graded A). Supervisors: Dr Maleeha Azam and Dr Muhammad Umar Khan.

PUBLICATIONS AND RESEARCH OUTPUTS

Ahmad, H., Khan, M.U., & Azam, M. (2024). Comparative Analysis of Machine Learning Methods for Enhancing Sleep Efficiency and Prediction. *ICSMAI 2024* (Springer). DOI: 10.1007/978-3-031-66854-8_1

Software and data. FinLaw-UK implementation and a 110-item UK financial-regulation QA benchmark (80 factual questions, 20 document tasks, and 10 case scenarios), publicly released: Finlaw-UK

RESEARCH EXPERIENCE

University of Bradford — Jan 2025 - Sep 2025

Research Assistant, Machine Learning & LLMs — Bradford, UK

- Designed and evaluated **FinLaw-UK**, a hybrid architecture combining retrieval-augmented generation, a Neo4j knowledge graph and a locally served Mistral 7B model to reduce hallucination in UK financial-regulation question answering, grounded in the FCA Handbook, PRA Rulebook, FRC standards and statutory instruments.
- Built the retrieval pipeline: clause-level segmentation of regulatory text, Sentence Transformer embeddings, cross-encoder re-ranking, and graph-based citation verification that flags unsupported references as hallucinations.
- Evaluated with RAGAS on a custom 110-item benchmark, implementing a strict graph-based verification gate that successfully flagged and refused fabricated citations
- Developed reproducible experimental pipelines and analysed model interpretability using feature attribution techniques.

COMSATS University Islamabad — Jul 2023 - Jul 2024

Research Assistant, Data Science — Islamabad, Pakistan

- Benchmarked 11 classifiers for diabetes risk prediction on a 253,680-record CDC BRFSS dataset (14% positive class), comparing random over-sampling, SMOTE and ADASYN for class imbalance with resampling confined to the training folds; Random Forest performed best on ROC-AUC and sensitivity.
- Analysed 20+ demographic, lifestyle and clinical indicators, identifying age, general health, BMI, blood pressure and income as the strongest diabetes correlates, and deployed the model as a public risk-screening web app (React, Flask): DiabetesSense.
- Led a first-author comparative study of sleep-efficiency prediction across four regression models (best model Random Forest, R^2 0.86), published at Springer.

INDUSTRY EXPERIENCE

Outlyst Oct 2025 - Mar 2026
AI / Machine Learning Engineer Leeds, UK (Remote)

- Developed backend dialogue-flow logic and workflow automation for an AI voice-agent system (Retell AI, FastAPI), integrating LLM-based dialogue with production infrastructure across 2,100+ calls.
- Reduced average end-to-end response latency 54%, from 2.4s to 1.1s, through async callbacks, connection pooling and backend optimisation, a decisive factor in conversational systems where response delay degrades interaction quality.

TECHNICAL SKILLS

Machine Learning & NLP : LLMs, Retrieval-Augmented Generation (RAG), Sentence Transformers, cross-encoder re-ranking, RAGAS evaluation, predictive modelling, ensemble methods, PyTorch, TensorFlow, scikit-learn

Knowledge & Data : Neo4j knowledge graphs, vector embeddings, semantic search, PostgreSQL, pandas, NumPy

Programming & Tools : Python, C++, SQL, TypeScript, FastAPI, Docker, Git, Linux, Ollama

LANGUAGES

English (IELTS 7.0, CEFR C1) | Urdu (Native) | German (A1.2)

CURRENTLY: Quality Control Assistant, Myton Food Group (Morrisons), Bradford, UK Apr 2026 - Present
Full-time role held while pursuing PhD applications and AI/ML research positions.

REFEREES

Dr Irfan Mehmood, Associate Professor in Business Analytics and Artificial Intelligence, University of Bradford
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Professor Tillal Eldabi, Professor of Business Analytics, University of Bradford | `t.a.eldabi@bradford.ac.uk` *(further referees available on request)*